PROGRESS REPORT FOR SPARK INTERNSHIP

- **Report Type**: Final
- **SPARK ID**: SPA222725
- **Name of intern**: Utkarsh Bajpai
- **Institute**: IISER Mohali
- **Name and department of assigned supervisor**: Dr. Puneet Jain, Dept. of Physics
- **Starting date of internship (20/05/2021)**:
- **Proposed End date of internship (25/07/2021)**:
- **Is there any break taken during the internship, e.g. due to the academic schedules of your parent institute?** No
- **Summary of work done. Use additional sheets if necessary. [Attached Below]**

<u>To be filled by the supervisor</u>

- How will you rate performance of the candidate in a scale of 0-10 (0 is poor, 10 is outstanding)? Add comments on the SPARK Intern performance.

Design of Low-Energy Thermionic Electron Gun

Project Report

SPARK
IIT Roorkee

Student
Utkarsh Bajpai

Supervisor
Dr. Puneet Jain

July 23, 2021

Acknowledgements

I sincerely thank Prof. Puneet Jain for giving me an opportunity to work on this project, his continuing support, and for multiple fruitful and enlightening interactions. Although I might not have been at my best productivity, these discussions greatly encouraged me from time to time and helped me in keeping myself motivated during these difficult times due to the pandemic.The disussions also helped in getting a better theoretical knowledge and practical skills and helped me in clearing any doubts in the project.

I would also like to express my gratitude to Mr Manish Vishwakarma, for providing assistance and excellent resources that helped in learning COMSOL.

Last but not the least, I would like to express my gratitude to the entire SPARK team, for providing such an excellent platform for getting hands on research experiences.

Utkarsh Bajpai
IISER Mohali
July 23, 2021

Abstract

In this project we first tried to understand the basic of electron beams and also the challenges to create a beam of high energy electrons and how to neutralize the effect of Coulomb-Coulomb repulsions using Pierce Geometry.Having obtained the basic theory we explored various COMSOL tutorial's to simulate a low energy electron gun in COMSOL.We also outlined the conceptual knowledge and tasks that we will try to explore in the future.

Contents

1 Introduction **5**

2 Theory **5**
2.1 Basics 5
2.2 Space Charge Limited Emission 6
2.3 Derivation of Child Langmuir Law in one dimension . 6
2.4 Deriviation of the angle for Pierce Gun 7
2.5 Finite Element Method 9

3 COMSOL Multiphysics **9**
3.1 Charged Particle Tracing 9
3.1.1 Particle Field Interaction, Non Relativistic . . 10
3.1.2 Bidirectional Particle-Field Interaction 10
3.2 Gun Design using COMSOL 10
3.3 Limitatons 12

4 Results **12**

5 Challenges and Future Outlook **13**

References **15**

1 Introduction

Due to scarcity in the drinking water due to various environmental and man made factors, importance of availability of pure water has tremendously increased.High energy electron beams cause the decomposition of impurity radicals and make their separation easier. In this project we learned about the basics of electron beam physics and also tried understand the basics of electron gun design using COMSOL Multiphysics software.

2 Theory

In his project we learned about various concepts related to electron ion beam.We also learned about various figures of merits(FoM) such as Current, Charge and Emittance. These are used to assess the beam quality.

2.1 Basics

The current inside the beam is convective current and the corresponding current density is,

$$\vec{J}_{conv} = \rho_v \vec{v}$$

Qualitatively, the term emittance at a point represents the divergence of the beam from the ideal orbit, at that point.Mathematically, it is the area of the beam in trace space plane, $[x, x']$

Beam brightness is the current density per unit solid angle in the axial direction. Brightness quantifies the maximum focused power flux of a beam.

The term Perveance(P) measures the magnitude of the space charge effects in the beams, it only depends upon the geometry of the extractor gap and the type of the particle.

$$P = I/{V_0}^{3/2}$$

The Child's Law puts a contraint on the maximum current density that can be obtained from the beam.For 1 dimensional case, the child law states that

$$j_{max} = \frac{4\epsilon_0}{9}\sqrt{\frac{2Ze}{m_0}}\frac{V_0^{\frac{3}{2}}}{d^2}$$

2.2 Space Charge Limited Emission

In most of the particle beams the electrons being emitted from the cathode have a low kinetic energy as a result the space charge(average effects of all the Coulomb interactions) density is very high near the cathode. These accumulated electrons create a potential barrier that makes electrons below a certain energy to revert back to cathode.

A virtual cathode is set up at a small distance from the cathode from which the emission of electrons take place.The distance of this virtual cathode is given by the Debye Length, $\lambda_d = \sqrt{\frac{kT_e\epsilon_o}{e^2 n_o}}$

2.3 Derivation of Child Langmuir Law in one dimension

Let us consider a parallel plate capacitor with having a potential difference of V_0 between the plates,the distance between the plates is a.

The Poisson equation for potential gives,

$$\frac{\partial^2 V}{\partial x^2} = -\frac{\rho}{\epsilon_o}$$

Consider a surface element $d\omega$ parallel to the YZ plane at some distance x from the cathode. The number of electrons to enter this element $d\omega$ in time dt are

$$dN = Jd\omega dt$$

where, J is the current density
The volume ocuupied by these electrons will be

$$dV = vd\omega dt$$

where v is the speed of the electrons.
The space charge density in this volume element will be,

$$\rho = \frac{dN}{dV} = \frac{J}{v}$$

For non relativistic speeds we have,

$$\frac{1}{2}mv^2 = eV_0$$
$$v = \sqrt{\frac{2eV}{m_e}}$$

The Space Charge density now becomes,

$$\rho = J\sqrt{\frac{m_e}{2eV}}$$

The Poisson equation takes the form,

$$\frac{\partial^2 V}{\partial x^2} = \frac{J}{\epsilon_0}\sqrt{\frac{m_e}{2eV}}$$

The boundary conditions are
V(0) = 0 V(a) =V_0 and $\frac{dV}{dx}|_{x=0} = 0$
The third boundary condition is due to the property of the curve of potential in case of Space Charge Limited flow. Solving the Poisson equation we get,

$$V = V_0\frac{x}{a}^{4/3}$$

Substituing this potential back into the poisson equation we get,

$$\frac{4}{9}\frac{V_0}{a^2}\frac{x}{a}^{-2/3} = \frac{J}{\epsilon_0}\sqrt{\frac{m_e}{2eV_0}}\frac{x}{a}^{-2/3}$$

Solving for J we get the Child Langmuir law

$$J = \frac{4\epsilon_0}{9}\sqrt{\frac{2e}{m_e}}\frac{V_0^{3/2}}{a^2}$$

2.4 Deriviation of the angle for Pierce Gun

In order to counteract the Coulomb repulsions between the electrons we need to design focusing electrodes(Pierce electrode).The appropriate angle with respect to the axis is calculated below,
The potential in the extractor gap varies as,

$$\phi(x, y, z) = V_0\left(\frac{z}{d}\right)^{\frac{4}{3}}$$

Here, d is the width of the extractor gap.
Let us consider a complex number,

$$C = z + \iota x$$

We know that any analytical function of complex variable C, f(C) satisfies the laplace equation, hence

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$$

The analytical function,

$$f(C) = V_0 \Big(\frac{u}{d}\Big)^{\frac{4}{3}} = V_0 \Big(\frac{z + \iota x}{d}\Big)^{\frac{4}{3}}$$

The electrostatic potential now becomes,

$$\phi(x, z) = V_0 \text{ Re } \Big(\frac{z + \iota x}{d}\Big)^{\frac{4}{3}}$$

In polar coordinates, $z = \rho \cos\theta$ and $x = \rho \sin\theta$

$$e^{\iota\theta} = \cos\theta + \sin\theta$$

Now, the potential becomes

$$\phi = V_0 \Big(\frac{\rho}{d}\Big)^{\frac{4}{3}} \text{ Re } (e^{\iota\theta})^{\frac{4}{3}}$$

The boundary conditions imply that potential at source, $\phi_1 = 0$ and potential at extractor is V_0, $\phi_2 = V_0$. At the source for $\phi = 0$, the argument of $e^{\frac{4}{3}\iota\theta}$ has to be $\pi/2$.

$$\frac{4}{3}\theta = \pi/2$$

$$\theta = \frac{3}{8}\pi \text{ rad } = 67.5^\circ$$

Thus, the focusing electrode should be inclined at angle 67.5° from the axis.

2.5 Finite Element Method

Most of the Numerical Simulation softwares use Finite Element Methods to solve the Poisson Equation and trace the particles.The key to Finite element method is that differentiation with respect to time is discretisized i.e, dt becomes Δt and the equations are solved in steps of Δt.As we reduce the Δt we get better and better results but at the cost of computational times.

The basic computations steps used by ray tracing softwares can be summarised as,

1. Solving the Poisson Equation, without including space charge to find the potential at all the points.
2. Calculating the electric fields from the potentials and advancing the particle orbits $[\vec{x}, \vec{v_x}]$.
3. Assigning beam space charge to all the mesh points using the particle trajectory derived earlier.
4. Re-Solving the Poisson equation, this time with space charge included.

   $$V(i,j) = \frac{1}{4}(V(i+1,j)+V(i-1,j)+V(i,j-1)+V(i,j+1))+\frac{\rho\Delta^3}{\epsilon_0}$$

5. The steps (2), (3) and (4) are iterated through until the ray reaches its final state.

3 COMSOL Multiphysics

COMSOL Multiphysics is a simulation software used to study coupled physical phenomenon through computer simulations and also to study multiple interacting physical properties.

3.1 Charged Particle Tracing

The Charged Particle Tracing module is primarily used to compute the paths of electrons, ions under the influence of electric and magnetic fields.

3.1.1 Particle Field Interaction, Non Relativistic

This interface is used to combine charged particle tracing with electrostatics.It is employed to model beams moving at non relativistic speeds that can contribute to the space charge density around them and in turn affect their own trajectories. The equation of particles in Non Relativistic domain can be described as

$$\frac{d(m_e\vec{v})}{dt} = e\vec{E} + e(\vec{v} \times \vec{B})$$

The force due to self magnetic fields can be neglected if the particles are moving with non relativistic speeds.

3.1.2 Bidirectional Particle-Field Interaction

When the density of electrons to be traced is high then Coulomb-Coulomb forces effects dominate, whereby each electrons affects the trajectory of every other electron.Bidirectional coupling involves a double side interactions between particles and fields.The electrons are allowed to contribute to the space charge and then this modified space charge density is used to find the potential and hence the force on the particles.

3.2 Gun Design using COMSOL

1. Including the relevant physics modules: The first step in gun modelling involves adding the related Physics interfaces to our model, for the case of our gun we used the `Particle Field Interaction, Non Relativistic` since energy of electron in our case is less than rest mass energy of electron(0.5 MeV).We also included the preset study `Bidirectionally Coupled Particle Tracing` [3.1.2].

2. Setting up Global Parameters: Having included the basic physics interfaces we start setting the geometry parameters for the gun which include shape and size of cathode, cathode-anode gap, anode shape, angle of cathode.

3. Building the Gun Geometry: We build the gun geometry using basic shapes like rectangle, polygon and a parametric curve to define the anode shape.we converted the whole sketch into a solid.

4. We then specified the particle type - electron using particle properties and also specified that the External Electric field should be contributed due to Electrostatics and Charge conservation nodes.

5. Setting up Space Charge Limited Emission: From the Multiphysics Couplings we select the cathode in our geometry and enable space charge limited emission.

6. Setting up the potentials: Using the Electrostatics node we set up the potentials at the cathode(ground) and the anode(75kV).

7. Defining Mesh: In COMSOL particles do no directly interact with the model geometry instead a mesh is used to mediate the geometry effects on the particles. For our gun, we used the `Physics Controlled Mesh` whereby the software automatically designs a mesh based on geometry.

8. Setting Beam Current: We used the `Particle Beam` node to set the beam current (1mA) and also specified the number of model particle per release(N) 10^4.Each model particle represents a number of real electrons, each contributing to the space charge density in the mesh cells through which the particles have passed.The frequency of the release of particles is given by,
$$f_{rel} = \frac{I}{|e|\,N}$$

9. Setting the Study: The bidirectionally coupled particle tracing alternatively calculates potential(stationary) and particle trajectories(transient).We set the time units of computation to be nanoseconds and the output times.We also specify the number of iterations(20).

10. Plots: We obtained the Particle Trajectory plots from the `2D Plot Group`. The trace space plots in the transverse direction X and Y wre also plotted using the `Phase Potrait` from `2D Plot Group`.

3.3 Limitatons

We try to estimate the number density of electrons in the electron beam that we tried to mode, For electrons in non relativistic domain,

$$E = \frac{1}{2}mv^2 = 75\text{keV} \tag{1}$$

$$v = 1.62 \times 10^8 \text{m/s} \tag{2}$$

The radius of our beam is 1 cm.

$$J = \rho v \tag{3}$$

$$J = nev \tag{4}$$

Multiplying by Area on both sides we get.

$$I = neV \times \pi(0.01)^2 \tag{5}$$

Out the values of I = 1mA, e = 1.6×10^{-19} and v from (2) we have,

$$n = 1.22 \times 10^{10} \frac{\text{particles}}{m^3}$$

Thus, the number of electrons in beam will be atleast the order of 10^8 to 10^9, while the model particles which we choose for computation is 10^4.So what essential is happening that we have reduced the effect of Coulomb-Coulomb repulsions by atleast around 10^6 times.This has greatly affected the results obtained, as now the software considers the particle rays to be independent of each other.

The main reason to use a low number(10^4) for model particles is due to our computational limitations.Since we were working in online mode,we were using a laptop with 8Gb RAM and `Intel Core i5(10 Gen) processor`.In future we look forward to increase the number of macro particles and obtain better results by including Space Charge Effects.

4 Results

Having designed a gun using COMSOL we performed the study and and obtained the particle trajectories plot. We also gradually changed the angle of the cathode from the axis and observed that as

we decreased the angle from 67.5 degree(Pierce Solution) the beam waist distance from the cathode decreases.

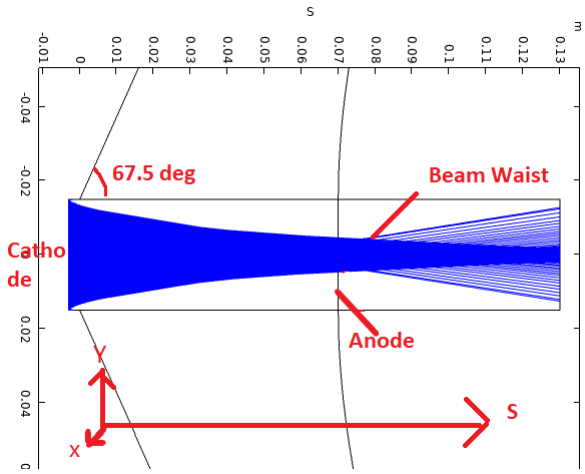
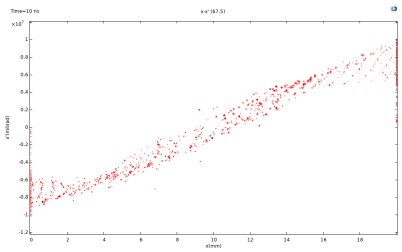
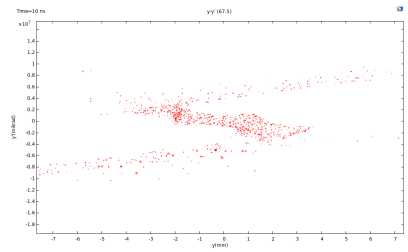


Figure 1: Particle trajectory for Pierce Angle(67.5) degree

We also plotted the trace space diagrams in the transverse directions X and Y.The plots are also shown below.We will further explore how to find the area of the trace space ellipse and thereby find the beam emittance.



(a) x-x'



(b) Y-Y'

Figure 2: Trace Space Plots

5 Challenges and Future Outlook

Working in online mode, there were various challenges that we faced in this experiment.First of all we had to reduce the current from 1 Ampere to 1 milli Ampere due to the computational times involved. we also had to keep the macro particles to 10^4 instead around 10^{10}

which led to ignorance of otherwise.space charge effects.

Nevertheless, we gained a lot of understanding of basic concepts of beam designs and insight into using COMSOL to model electron guns.We will further exlore various other concepts in the future like find emittance from trace space plots, having deep understanding of gun geometries and implementing the same in COMSOL. Having done that, we will try to optimise the electron to be used for societal applications. We also look forward to pursue further long-term projects and higher studies in the field of Accelerator Physics

References

[1] Modeling an pierce electron gun in comsol. https://www.comsol.com/blogs/modeling-a-pierce-electron-gun-in-comsol-multiphysics/. Accessed: 2021-07-2.

[2] W B Herrmannsfeldt. Egun: An electron optics and gun design program. 10 1988.

[3] W. B. Herrmannsfeldt. Developments in electron gun simulation. *Phys. Scripta T*, 71:28–33, 1997.

[4] Jr. Humphries, Stanley. *Charged particle beams*. 1990.

[5] P. Jánský, Bohumila Lencová, and Jakub Zlamal. Accurate calculations of thermionic electron gun properties. 01 2008.

[6] Alberto Leggieri, Davide Passi, Franco Di Paolo, Bruno Spataro, and Egor Dyunin. Advanced design of a low energy electron source. In *2015 IEEE MTT-S International Conference on Numerical Electromagnetic and Multiphysics Modeling and Optimization (NEMO)*, pages 1–4. IEEE, 2015.

[7] Gerard Tranquille and Jean Cenede. Design and Optimisation of the ELENA Electron Cooler Gun and Collector. page THPMB048. 3 p, 2016.